FRANCESCO BANELLI

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EDUCATION

ETH Zurich *Sept 2024 –May 2027 (expected)*
MSc in Robotics, Systems and Control GPA: 5.83/6.0
Master's thesis at Disney Research, starting Nov 2026

Politecnico di Milano *Sept 2021 –Jul 2024*
BSc in Automation and Control Engineering GPA: 29.35/30
Graduation Grade: 110/110 cum Laude

RESEARCH EXPERIENCE

AI Engineering Intern, Flexion Robotics - Perception Team *Mar 2026 –Oct 2026*
Startup building robot-agnostic intelligence for humanoids. Supervisors: Julian Nubert, Dylan Vogel.

Led an exploratory project on proprioceptive state estimation for humanoids, aimed at the layer of the estimation stack that must never fail. I developed a learning-augmented estimator built on the principle that a neural network should model only what the kinematics cannot.

- Showed that it generalizes to unseen motions, where fully learned methods fail.
- Deployed on the Unitree G1 and LimX Oli; outperforms classical estimators on every terrain.
- Continued after the internship as first author of a paper for IEEE RA-L.

Student Researcher, ETH Zurich - Raffaello D'Andrea's group *Feb 2025 –Feb 2026*
Semester project and continued research. Supervisors: Antonio Terpin, Raffaello D'Andrea.

Worked on fast and accurate fluid-flow estimation: the feedback needed to train reinforcement-learning policies for active flow control directly in the real world.

- Developed ART of PIV, a router that learns to dynamically select the best flow estimator for each frame. Deployed it on the group's tabletop water channel.
- Co-authored a consensus-ADMM method that fuses multiple flow estimates per pixel (CDC 2026).
- Co-developed Flow Gym, a JAX toolkit for flow estimation (SoftwareX), and lead its development.
- Co-developed SynthPix, a GPU-parallel generator of synthetic PIV data for training at scale.

Student Researcher, Politecnico di Milano - Nicola Gatti's group *Nov 2023 –Jul 2024*
Supervisor: Pierriccardo Olivieri.

- Studied multi-agent coordination and distributed optimization for drone formation control.
- Implemented distributed MPC and potential-field controllers, comparing speed and stability.

PUBLICATIONS

PIVOT: Humanoid State Estimation with Learned Trust *In preparation, IEEE RA-L*
Francesco Banelli, Dylan Vogel, Julian Nubert

ART of PIV: Agentic Real-Time Particle Image Velocimetry for Active Flow Control
Antonio Terpin*, **Francesco Banelli***, Alan Bonomi*, Raffaello D'Andrea *In preparation*

PIV Refinement via Consensus ADMM *CDC 2026*
Alan Bonomi*, **Francesco Banelli***, Antonio Terpin*

Flow Gym *SoftwareX*
Francesco Banelli[*], Antonio Terpin[*], Alan Bonomi, Raffaello D'Andrea

SynthPix: A lightspeed PIV image generator *SoftwareX*
Antonio Terpin[*], Alan Bonomi[*], **Francesco Banelli**, Raffaello D'Andrea

[*]Equal contribution

HONORS AND AWARDS

Merit Scholarship for Out-of-Town Students, Politecnico di Milano *2021–2024*
Awarded 3 consecutive years in a competitive ranking

2nd Place Nationally, Matematica & Realtà Modeling Competition (2×) *2019, 2021*
Individual mathematical modeling competition; listed in the Italian National Registry of Excellence

TEACHING

Teaching Assistant, ETH Zurich *Sept 2025 –Present*
Game Theory (2 semesters), Linear System Theory (1 semester)

Peer Tutor, Politecnico di Milano *Feb 2023 –Jun 2024*
Linear Algebra, Calculus I, Physics, Control Systems (6 students)

LEADERSHIP AND ENTREPRENEURSHIP

Visual Odometry Team Lead, AEA - Politecnico di Milano *May 2023 –Dec 2023*
Automation Engineering Association, developing a drone built entirely by students.

- Led a three-student team working on visual odometry and sensor fusion.

Co-founder & CEO, CapaCity *Sept 2022 –Nov 2023*
Startup project helping shared-micromobility operators move from free-floating to station-based fleets.

- Selected among the top 45 projects of Switch2Product, Politecnico di Milano's accelerator.
- Won the Verde e Blu Challenge, pitching the project to a public audience.